



Technical Test

Dear candidate,

As part of the recruitment process at Bagira Systems, you are required to complete this assignment. You will receive the assignment at a specified time (e.g., 9 AM) and must submit your solution within 24 hours (e.g., by 9 AM the next day), ***please read the description carefully before starting.***

Objectives:

Implement missing parts in a provided code skeleton (backend and frontend) for a small web application used to create training scenarios and manage entities inside each scenario

Instruction:

1. Clone this repository – [Bagira-Assignment](#) and create a fork with the following structure – ‘BagiraAssignment_<YourName>’
2. The Repo contains frontend skeleton for React and Angular implementation – choose one to implement and specify which one you chose.
3. Follow the **README** instructions on the backend and frontend folders for detailed instructions.
4. Implement the missing functionality marked in the code (TODOs) and ensure the app works end-to-end.

Requirements:

1. Data rules:
 - EntityType must be one of the following - Soldier, Tank, Drone, Aircraft, Vehicle, Civilian.
 - TaskForce must be one of the following - Friendly, Enemy.
 - Latitude must be between -90 and 90.
 - Longitude must be between -180 and 180.
 - Entities always belong to exactly one Scenario (no orphan entities).
2. Screens:
 - Scenario List – view all scenarios, navigate to details, create.
 - Create Scenario – the form should contain name and description.
 - Scenario Details – view scenario info and manage its entities.
 - Create Entity – the form must include:
 - EntityType
 - TaskForce
 - Name/Callsign
 - Latitude
 - Longitude
3. Frontend Features:
 - Filtering & Sorting - scenarios and entities tables.
 - Search - global search bar navigation to search scenarios and entities by names
 - Data Visualization:
 - i. Map view for entities (using coordinates)
 - ii. Toggle between table view and map view
 - iii. Display entities on map with Type/TaskForce filtering
 - iv. Click entity on map to see details
 - UX Polish:
 - i. Toast notifications for success/error operations
 - ii. Enhanced loading states (skeleton loaders, button spinners)
 - iii. Real-time form validation feedback
 - Validation - Frontend must validate:
 - i. Latitude must be between -90 and 90
 - ii. Longitude must be between -180 and 180
 - iii. Entity types are fixed: Soldier, Tank, Drone, Aircraft, Vehicle, Civilian



- iv. TaskForce must be Friendly or Enemy
 - v. Required fields (name, etc.)
- 4. Docker:
 - After implementing the core functionality, containerize the solution using Docker:
 - i. Create Docker files for Backend and Frontend
 - ii. Create a docker-compose.yml.
 - iii. Ensure the solution runs with: docker-compose up
 - iv. Update the main README with Docker instructions
 - Notes:
 - i. Use multi-stage builds for frontend (build + serve)
 - ii. Configure CORS in backend to allow frontend origin
 - iii. Use environment variables for API base URL in frontend
 - iv. Document any required environment variables
- 5. Data persistency
 - **Bonus:** If you have time, connect to a DB of your choosing
- 6. Optional bonus
 - Add basic tests for validation (backend) or key UI flows (frontend).
 - Implement Update and Delete capabilities end-to-end for Scenarios and Entities controller.

Acceptance criteria

- User can create a scenario and see it in the scenario list.
- User can open scenario details and view entities belonging to that scenario only.
- User can select TaskForce (Friendly/Enemy) when creating entities, and TaskForce is displayed in entity lists.
- User can filter and sort entities by Type, TaskForce, and other fields.
- User can search for scenarios and entities.
- User can view entities on a map and toggle between table and map views.
- Toast notifications appear for successful operations and errors.
- Frontend validates all input fields (coordinates, types, required fields).
- Solution runs locally.
- Solution runs in Docker using docker-compose up with all services properly configured and communicating.
- Using clean architecture.
- Code quality and maintainability (structure, naming, separation of concerns).
- Validation and error handling.
- Reasonable scope management within the timebox.

Please provide:

- A link to your branch.
- Short notes (bullet points) describing what you completed and any trade-offs.

Good Luck!